

News Media Coverage Predictors of Valley Fever Case Rates

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Abstract. Endemic to California’s Central Valley and southwestern parts of the United States, Valley Fever is a respiratory disease that may be influenced by various environmental factors. Since the disease is prevalent in areas under arid climate conditions, prior research focuses on how environmental conditions predict future case reports. However, there is limited research on how public awareness and news media coverage plays a role in these predictions. With Fresno and Kern County being high-incidence regions in the Central Valley, this project examines how Google News articles mentioning terms related to Valley Fever improve predictive modeling of monthly case rates in these counties. To evaluate predictive accuracy, models such as baseline regression, Long Short-Term Memory, and Seasonal Auto Regressive Integrated Moving Average models are compared. The results suggest that news articles on Valley Fever that contain risk-related terminology and regional mentions of Central Valley counties may influence reported case patterns across Fresno and Kern County.

Keywords: Valley Fever · News Media Analysis · Natural Language Processing · Long Short-Term Memory · Time-Series Forecasting

1 Introduction

1.1 Valley Fever as a Public Health Concern

Coccidioidomycosis, also known as Valley Fever, is a fungal disease that has become a growing public health concern for the Southwestern parts of the U.S. It is most prevalent in semi-arid regions where there is a severe lack of available water, dusty winds, and dry conditions [1]. The disease accompanies these conditions because fungal spores can become airborne when dry soil is disturbed, which then increases the risk of inhalation and infection [1]. Although it cannot be contracted interpersonally, it is still a public health concern due to increasing Valley Fever incidence rates and the spread of the disease beyond historically endemic areas [1].

Recent research suggests that that climate change may further contribute to the spread of Valley Fever. In a recent study by Pawan Kumar et al., researchers

found that increased temperatures and prolonged dry conditions due to climate change may increase the frequency of dust storms and soil disturbance events [1]. These events allow fungal spores to become airborne through the wind and transported across larger geographic regions outside of just the Central Valley [1]. This exemplifies how environmental and regional conditions may influence Valley Fever exposure risk, which motivates the use of environmental predictors within forecasting models for future case rates.

With the environment playing a huge factor in contracting Valley Fever, prior research has incorporated environmental factors such as precipitation, wind, and air quality into time-series forecasting models to predict case rates of the disease [2]. In a research paper by Nathalie Valenzuela and Mario Bañuelos, the role of California wildfires in predicting case rates was explored using deep learning approaches such as Long Short-Term Memory (LSTM) models. Wildfire acres burned was used as a predictor alongside other environmental variables, which added predictive power to these models [2]. This highlights the significance of integrating machine learning frameworks into Valley Fever case rate forecasting in California’s Central Valley.

1.2 Public Awareness and Media Coverage

Beyond environmental conditions, it is also important to consider other external factors that may contribute to case reporting of the disease such as misdiagnosis. Since symptoms of the disease frequently resemble other respiratory illnesses such as influenza or COVID-19, it is often not reported initially as Valley Fever [3]. This may result in delays in diagnosis, which can skew case reporting timelines that lag behind the actual onset of symptoms. The delays and misdiagnosis may coincide with low patient awareness, as individuals who are not familiar with the disease are unlikely to prompt their physicians to test for it.

Previous survey research by Glorietta Hurd-Kendeti et al. shows that the lack patient awareness is widespread, with less than half of Californians during 2016 and 2017 stating that they had heard of Valley Fever. Moreover, only 25% of respondents living in high-incidence regions, such as Fresno and Kern County, were aware that they lived in areas where fungal spores exist [4]. This demonstrates a large awareness gap and motivates the question on whether public awareness may play a role in reporting.

Since there is limited research on whether awareness through news media coverage on Valley Fever contributes to case reporting, we explore the relationship in this study. Specifically, we scrape Google News articles mentioning terms related to Valley Fever and extract monthly media features such as news article count, regional mentions, and risk-related terminology. Looking at the content within each news article allows us to evaluate how the severity of the disease is communicated to the public and whether media urgency corresponds to changes in reported case rates across Fresno and Kern County.

2 Methods

2.1 Valley Fever Case Rate Data

Monthly Valley Fever case counts in Central Valley counties were converted to case rates based on the number of cases per 100,000 people. The case rates were calculated using population estimates provided by U.S. Census Intercensal data and were preprocessed in a ValleyCast GitHub repository [5]. Within this repository, multiple environmental datasets were aggregated into county-level monthly observations. Environmental features include precipitation, wind metrics, Air Quality Index (AQI) Particulate Matter (PM) 2.5 and 10, earthquake totals, fungicide application totals, and wildfire acres burned. Due to Valley Fever's high-incidence regions being Fresno and Kern County, this study will center data that ranges from dates starting at the beginning of 2008 to the end of 2015 within these counties.

2.2 Google News Feature Engineering

To explore Google News media coverage on Valley Fever, the Python module `pygooglenews` [6] was used to query the number of monthly news articles mentioning terms such as "Valley Fever", "Coccidioidomycosis", and "San Joaquin Valley Fever." Since media sources frequently refer to the disease using geographic naming associated with California's Central Valley, the last search term was included to improve article retrieval.

Beyond the monthly article count, the full text from each article was extracted using the `newspaper4k` [7] Python module. Natural Language Processing techniques were used to normalize article text by removing URLs, bracket content, parentheses with citations, external source names, and white space. After this preprocessing pipeline, keyword extraction was used to look at regional mentions and risk-related language associated with Valley Fever reporting. Specifically, the number of times a Central Valley county such as Fresno, Kern, Tulare, Kings, Madera, Merced, and articles containing "Central Valley" were scraped. Moreover, the number of times a risk word such as "outbreak", "surge", "spike", "increase", "danger", "severe", "deadly", "hopsital", or "death" were also scraped.

Risk word and regional mention frequencies were converted to rates by dividing the mention frequency over the number of articles. These media-related variables were aggregated by year and month to align with the monthly Valley Fever case rate and environmental data. To investigate whether Valley Fever case rates respond to news coverage from months prior, article volume, regional mention rates, and risk-related mention rates were time-lagged by different months. Since three months had the strongest overall model performance, a time-lag of three months was chosen.

2.3 Regression and Influence Diagnostics

Multiple linear regression models were used to evaluate the relationship of Valley Fever case rates at month t and lagged news media variables at month $t - 3$. The response variable was defined as the log-transformed monthly case rate and predictor variables as the time-lagged media variables, seasonal effects, and county interaction terms. Log-transforming the response variable helped achieve model assumptions of normality and constant variance. Since the monthly news media data is equivalent across both Fresno and Kern County, the standard errors in the model were clustered by month. This allows for observations within the same month to share correlated error structure and display more reliable regression results.

Cook's Distance (D_i) and Difference of Fits ($DFFITs_i$) influence diagnostics were used to identify influential observations that disproportionately affect the regression model. Separate models were refit after removing influential observations exceeding the DFFITS threshold,

$$|DFFITs_i| > 2\sqrt{\frac{k+2}{n-k-2}}$$

and Cook's Distance threshold,

$$D_i > \frac{4}{n},$$

where k denotes the number of predictor terms and n denotes the total number of observations [8]. Comparing models helped determine whether estimated effects of news media variables remained consistent across different influence diagnostic methods.

2.4 LSTM Time-Series Forecasting

Long Short-Term Memory (LSTM) is a neural network that is used to learn patterns in either the short or long term of past sequential data to predict future terms [9]. In contrast to traditional neural networks that treat observations independently, LSTM models retain information from previous time steps with the use of memory cells and gating mechanisms. These gates decide which information is remembered, updated, or forgotten over time [9]. It allows the model to capture both short-term and long-term temporal dependencies within sequential observations such as time-series data [9]. Since Valley Fever case rates in Fresno and Kern County have seasonal outbreak patterns, LSTM models were chosen to capture temporal relationships between historical case rates, environmental variables, and news media predictors.

The inputs for this study include environmental variables from the ValleyCast GitHub repository [5] and scraped news media predictors, with log-transformed Valley Fever case rates being the target variable. LSTM models were trained for Fresno and Kern county using hyperparameters such as sequence length, hidden

size, dropout, learning rate, and number layers that were fine-tuned from a grid search. An 85%/15% time-chronological train/test split of the data was used to evaluate forecasting performance on unseen future data.

The Root Mean Squared Error (RMSE) for each split was also analyzed to evaluate predictive performance, with an emphasis on achieving a low test RMSE to avoid overfitting. Moreover, time-series visualizations were used to compare the predicted and observed Valley Fever case rates across both splits. Permutation Feature Importance (PFI) helped evaluate how much each individual predictor variable contributed to LSTM forecasting performance. PFI scores are computed by randomly shuffling the values of each feature’s column and calculating the increase in RMSE after the shuffle [10]. Features with a higher RMSE increase indicate that the model relied more on the feature [10]. Since the PFI scores vary across multiple LSTM runs, the average PFI scores across 100 runs were calculated for both counties to reduce variability caused by model training.

2.5 SARIMAX Time-Series Forecasting

Seasonal AutoRegressive Integrated Moving Average with Exogenous Variables (SARIMAX) models build off traditional ARIMA forecasting models by specifying a seasonal structure and considering predictor variables that exogenously influence the response variable [11]. The autoregressive (AR) component incorporates the temporal dependence between current Valley Fever case rates and previous monthly observations. Moreover, the moving average (MA) component captures relationships between current values and previous forecast errors [11]. Lastly, the integrated component ensures that the data is stationary by removing long-term trends and differencing data observations. [11]. These components correspond to parameters in the SARIMAX model, which is defined as $\text{SARIMAX}(p, d, q)(P, D, Q, s)$.

Table 1: SARIMAX model parameter definitions [11].

Parameter Definition	
p	Non-seasonal autoregressive order
d	Non-seasonal differencing order
q	Non-seasonal moving average order
P	Seasonal autoregressive order
D	Seasonal differencing order
Q	Seasonal moving average order
s	Seasonal period length

The difference between the non-seasonal and seasonal parameters are that the seasonal components model patterns that occur across patterns that repeat over a fixed time intervals [11]. Since Valley Fever case rates have strong yearly trends

that are exist alongside seasonal weather patterns, incorporating this structure in the model is important. In this study, a seasonal period length of $s = 12$ was used to represent yearly seasonality in monthly observations. Thus, the seasonal captures relationships across observations such as t , $t - 12$, $t - 24$, and other yearly lag intervals.

To determine other hyperparamters in the SARIMAX model, Partial Autocorrelation Function (PACF), Autocorrelation Function (ACF), and Augmented Dickey-Fuller (ADF) analyses were conducted. PACF plots were used to identify the autoregressive order p by analyzing the direct correlation between current observations and previous lagged values [11]. Similarly, ACF plots were used to determine the moving average order q by evaluating overall serial correlation across lagged observations [11]. Lastly, the ADF test was used to evaluate whether the data was stationary [11] and determine the differencing order d .

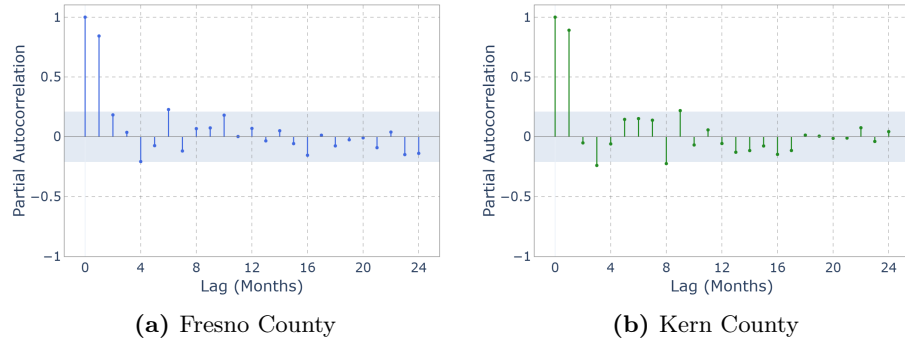


Fig. 1: Partial Autocorrelation Function (PACF) plots for monthly log-transformed Valley Fever case rates in (a) Fresno County and (b) Kern County.

The largest spikes in both county PACF plots occur at a time-lag of 1 and 2 months, suggesting a strong autoregressive dependence across recent observations of case rates. These spikes exist outside of the highlighted confidence interval, which also indicates that statistically significant partial autocorrelation occurs at lower lag values. Since the strongest direct relationship occurred at a time lag 1 month, an autoregressive order of $p = 1$ was chosen for the SARIMAX models.

In a similar ACF analysis, there were high early autocorrelation values and a decrease over larger lags. Since Fresno and Kern County showed these results, there was temporal correlation in monthly Valley Fever case rates. The captured relationship between current case rate values and previous forecast errors, from the ACF analysis resulted in the choice of a moving average order, $q = 1$ across both counties. Lastly, ADF testing showed that the log-transformed case rate data was not stationary and differencing was necessary for both counties, $d = 1$.

After conducting PACF, ACF, and ADF analyses, the final SARIMAX model was defined as SARIMAX(1, 1, 1)(1, 0, 1, 12). The seasonal differencing order was specified as $D = 0$ due to the yearly seasonal structure remaining relatively stable after non-seasonal differencing. SARIMAX models were used in this study to compare whether classical statistical forecasting or deep learning methods, such as LSTM modeling, capture trends in Valley Fever case rates better. Similar to the LSTM models, the forecasting performance of the SARIMAX models in this study were evaluated using RMSE values and time-series visualizations across training and test splits.

3 Results

The following results evaluate the relationship between environmental variables, news media features, and Valley Fever case rates across Fresno and Kern County.

3.1 News Media Trends

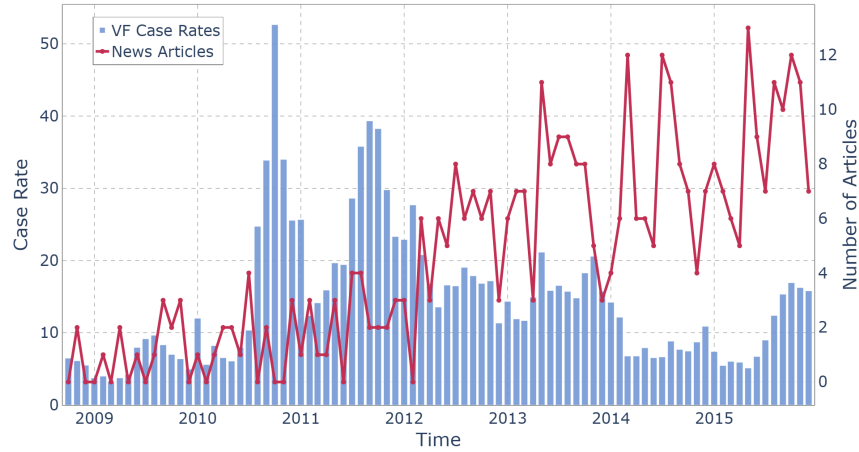


Fig. 2: Dual-axis chart of Kern County monthly Valley Fever (VF) case rates and news article volume from 2008 to 2016.

Valley Fever case rates in Kern County spiked around 2011 and 2012 before gradually declining in the following years. During this same time period, the number of Google News articles mentioning terms related to Valley Fever had a gradual increase. Overall, periods of elevated case rates were followed by higher levels of news media coverage, suggesting a delayed relationship. Although article frequency spikes do not completely align with each case rate spike, the delayed relationship motivated the choice of time-lagged news media variables within the regression and forecasting models.

3.2 Regression Results

Table 2: Regression coefficient estimates after removing influential observations using Cook’s Distance and Difference of Fits (DFFITS) thresholds.

Variable	Coefficient (Cook’s Distance)	Coefficient (DFFITS)
Intercept	2.0831*	2.0771*
(Number of Articles) $_{t-3}$	-0.0767*	-0.0772*
(Risk Mentions Rate) $_{t-3}$	-0.0474*	-0.0447*
(Cen Valley Mentions Rate) $_{t-3}$	-0.0341	-0.0382*
(News Articles) $_{t-3} \times$ Kern	0.0834*	0.0870*
Kern County	0.7673*	0.7345*
Spring Season	-0.3800*	-0.3885*
Summer Season	-0.1927	-0.1757
Winter Season	-0.1700	-0.2087
Observations	168	173
Adjusted R^2	0.644	0.623

Notes: Response variable was $\log(1 + \text{VFRate})_t$, using case rates from Fresno and Kern County, with standard errors clustered by month. Significance level: * $p < 0.05$.

Across both regression models, the significance levels and coefficient estimates remained relatively stable using both influence diagnostics, which suggests that these findings were robust to those influential observations that were removed. The negative coefficient estimates for the number of Google News articles on Valley Fever indicate that a higher article volume from three months prior was associated with lower log-transformed case rates of the current month. Similarly, the negative coefficients for the risk-related mentions rate and Central Valley mentions rate also indicate a negative association with case rates. However, the positive interaction coefficients between the number of articles and Kern County suggest that the relationship between article volume and case rates differed in Kern County relative to Fresno County.

Beyond the news media variable coefficient estimates, Kern County in general had higher log-transformed case rates in comparison to Fresno County. Moreover, the seasonal effects show that only Spring was statistically significant. The negative coefficient suggests that case rates during this season were lower in comparison to the baseline season, Fall. Overall, the adjusted R^2 values across both models indicate that more than 62% of the variation in the log-transformed case rates is explained by time-lagged news media variables, county interaction terms, and seasonal effects.

3.3 Forecasting Results

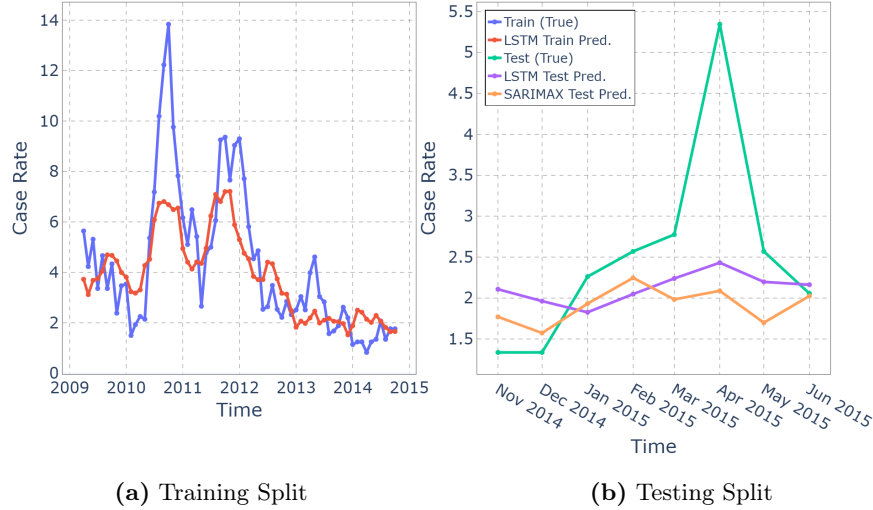


Fig. 3: Fresno County (a) training and (b) testing performance of the LSTM and SARIMAX models for Valley Fever case rate prediction.

The predictions for Fresno County from the LSTM model generally followed the overall temporal trends during the training split. However, the major outbreak spike during in 2010-2011 was somewhat underestimated. The predictions during the testing split varied, with the LSTM predictions having more variability than the SARIMAX predictions. The general magnitude of the LSTM testing predictions were also captured more well despite limited testing observations, which suggests that the LSTM model generalized to unseen data.

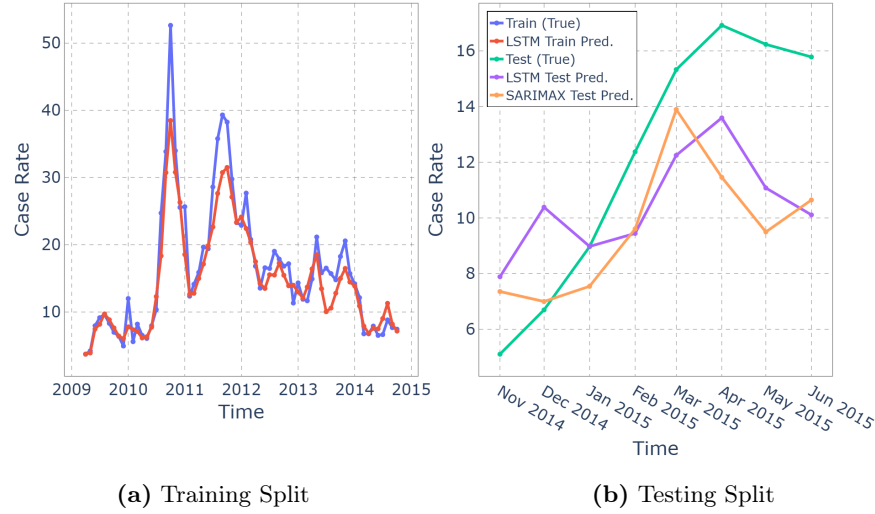


Fig. 4: Kern County (a) training and (b) testing performance of the LSTM and SARIMAX models for Valley Fever case rate prediction.

For Kern County, the training predictions followed the observed values more well than it did for Fresno County. The testing split showed more variability across both LSTM and SARIMAX model predictions, with the outbreak spike around April 2015 being captured reasonably well. Both models followed the general magnitude of the testing predictions, with slight underestimation.

Table 3: Forecasting performance comparison using RMSE values.

Fresno County		
Model	Train RMSE	Test RMSE
LSTM	1.81	1.13
SARIMAX	1.60	1.25
Kern County		
Model	Train RMSE	Test RMSE
LSTM	3.53	3.61
SARIMAX	5.31	4.83

Across Fresno and Kern County, the LSTM models had a lower test RMSE value, suggesting stronger forecasting performance during this split. However, the SARIMAX performed better during the training split for Valley Fever case

rate predictions in Fresno County. Overall, the LSTM model outperformed the SARIMAX model in both splits for Kern County.

3.4 Permutation Feature Importance

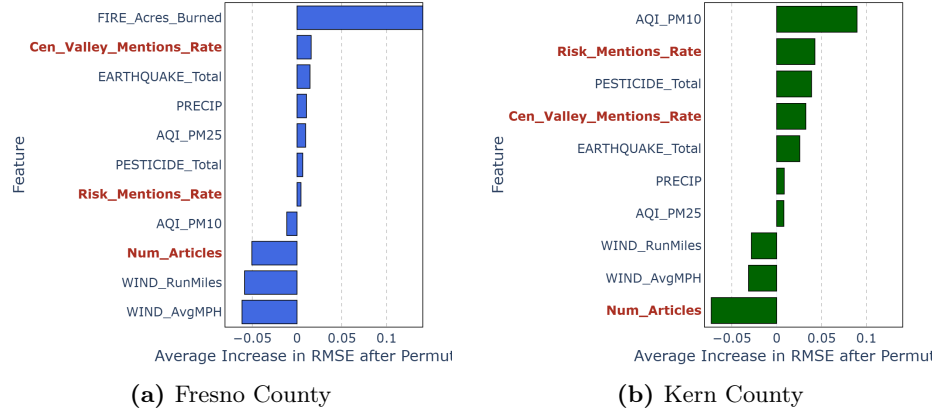


Fig. 5: Average permutation feature importance (PFI) scores across 100 LSTM runs for (a) Fresno County and (b) Kern County. Media-related variables are highlighted in red.

The PFI scores varied across Fresno and Kern County, with environmental variables such as wildfire acres burned and AQI PM 10 being the strongest predictors across both counties. The Central Valley mentions rate in Fresno County and risk word mentions rate in Kern County were one of the highest ranked PFI scores out of other environmental and news media variables. This suggests that these media variables contributed the second most to LSTM forecasting performance on average of 100 model runs. However, the number of Google News articles on Valley Fever had a negative PFI score across both counties, which may indicate correlated predictors or instability caused by feature permutation.

4 Discussion and Conclusion

4.1 Media Coverage and Model Implications

This project investigates whether news media coverage on Valley Fever contributes to case reporting of the disease by extracting Google News article frequencies, Central Valley county mentions, and risk-related terminology from these articles using Natural Language Processing techniques. Due to the high incidence rates in Fresno and Kern County, the case rates from these counties

were used as the target variable in baseline regression, LSTM, and SARIMAX models. Going beyond environmental features from prior research and adding news media features showed that media coverage may influence case rates across Fresno and Kern County.

The three-month time-lagged media variables from the regression models had a negative association with log-transformed case rates, which may suggest that coverage may act as a public health warning. When news coverage increases, people in Fresno and Kern County may take preventive measures such as wearing masks during dust storms, with risk-related terminology in these articles emphasizing the severity of the disease. However, this pattern could also reflect an overlap with seasonal cycles, where media coverage may peak during dry, dusty months and drop in the following subsequent Winter months. The seasonal effects from the regression models support this presence of a seasonal structure in Valley Fever case rates, with Spring having significantly lower case rates relative to Fall.

4.2 Seasonality and Forecasting Implications

Seasonality of Valley Fever outbreak patterns was captured in the LSTM and SARIMAX forecasting models using media-related variables from the regression models and environmental conditions from prior research. The SARIMAX framework provided more of a seasonal forecasting structure with less variability in the testing split, while LSTM models captured more nonlinear outbreak variability overall. Although the LSTM training RMSE value was higher in Fresno County, the LSTM model still outperformed SARIMAX overall. This suggests that LSTM modeling provides a modest improvement in predictive performance and SARIMAX remains a competitive classical forecasting method.

The PFI scores showed that environmental conditions remain critical to predicting monthly Valley Fever case rates, with the media-related variables such as Central Valley mention rates and risk-related terminology also contributing high predictive power to the LSTM models. The strong rankings of these variables suggest that the severity and media urgency communicated within Google News articles may correspond to changes in Valley Fever case rates across Fresno and Kern County. However, it's important to acknowledge that limiting the models to only two counties within California's Central Valley may reduce generalizability to other endemic regions. Moreover, Google News articles may not fully represent broader public awareness, since individuals may receive information from other media platforms, healthcare providers, or word-of-mouth.

4.3 Limitations and Future Work

With Valley Fever awareness being a broad topic, future work should expand this forecasting framework by incorporating additional awareness indicators such as Google Trends, social media activity or public health behavior measured by healthcare providers. More advanced Natural Language Processing approaches beyond regional mentions and risk-related terminology may also improve the

extraction of media urgency from news article content. The inclusion of other endemic regions of the Southwestern parts of the U.S. would also improve generalization for these models. Ultimately, this study demonstrates that combining environmental forecasting and news media analysis may improve the understanding of Valley Fever reporting patterns in Fresno and Kern County.

Acknowledgments. This material is based upon work supported by Dr. Mario Bañuelos, the MBG Research Group, and the California State University, Fresno Department of Mathematics.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Kumar, P., Jayan, J., Jena, D., Rai, N., Sah, S., Mawejje, E.: Valley fever on the rise: How climate change is fueling a public health crisis. *New Microbes and New Infections* **62**, 101500 (2024). <https://pmc.ncbi.nlm.nih.gov/articles/PMC11491959/>
2. Valenzuela, N., Bañuelos, M.: The Role of California Fires in Predicting Valley Fever. In: *Proceedings of the International Conference on Pattern Recognition*, pp. 1–10 (2023)
3. Centers for Disease Control and Prevention: Valley Fever (Coccidioidomycosis). <https://www.cdc.gov/valley-fever/about/index.html>,
4. Hurd-Kundeti, G., Sondermeyer Cooksey, G.L., Jain, S., Vugia, D.J.: Valley Fever (Coccidioidomycosis) Awareness — California, 2016–2017. *Morbidity and Mortality Weekly Report* **69**(42), 1512–1516 (2020). <https://doi.org/10.15585/mmwr.mm6942a2>
5. Bañuelos, M.: ValleyCast: Environmental and Valley Fever case rate aggregation pipeline. GitHub repository, <https://github.com/MBanuelos/ValleyCast>
6. Bugara, A.: pygooglenews: A Python package for Google News RSS feeds. GitHub repository, <https://github.com/kotartemiy/pygooglenews>
7. Paraschiv, A.: newspaper4k: Article scraping and curation for Python. GitHub repository, <https://github.com/AndyTheFactory/newspaper4k>
8. Pennsylvania State University: STAT 462: Applied Regression Analysis – Identifying Influential Data Points. <https://online.stat.psu.edu/stat462/node/173/>
9. Hochreiter, S., Schmidhuber, J.: Long Short-Term Memory. *Neural Computation* **9**(8), 1735–1780 (1997). <https://doi.org/10.1162/neco.1997.9.8.1735>
10. Altmann, A., Tolosi, L., Sander, O., Lengauer, T.: Permutation importance: A corrected feature importance measure. *Bioinformatics* **26**(10), 1340–1347 (2010). <https://doi.org/10.1093/bioinformatics/btq134>
11. Alharbi, F.R., Csala, D.: A Seasonal Autoregressive Integrated Moving Average with Exogenous Factors (SARIMAX) Forecasting Model-Based Time Series Approach. *Inventions* **7**(4), 94 (2022). <https://doi.org/10.3390/inventions7040094>